Probability

- introduction
  Prototype of Existing Phinomenat
- Probability :
- qualification of probable event

- Experiment :— outcome
  Run

- deterministic experiment
  2+2 = 4 unique outcome
  (remain same outcome every time)

- random experiment
  2+2 → the outcome is varys every time
  3 4 5 ~~uniqu~~ random

* List all possibilites :—
  sample space [S]

Lekhak 0918

Expt 1 :- Toss a coin
{ Head, Tail }

Expt 2 :- Toss a coin Twice, 2 Times

Toss 1	Toss 2
Head	Head
Head	Tail
Tail	Head
Tail	Tail

Head → H, Tail → T

$S = \{ (H.H), (H.T), (T.H), (T.T) \}$

Expt 3 :- Toss two coins simultaneously.

coin 1	coin 2
H	H
H	T
T	T
T	H

$S = \{ (H,H), (H,T), (T,T), (T,H) \}$

EVENT :- is a subset of sample space
denoted by [A, B, C, D, E]

Expt 4 :- Roll a die
$S = \{ 1, 2, 3, 4, 5, 6 \}$
A = getting an **odd** number
$\{ 1, 3, 5 \}$
B = getting a no multiple of 3
$\{ 3, 6 \}$
C = getting a no 7
$\{ \phi \}$

S → sample space
Event : A, B, C, D, E

- equally likely events. - event same chance of occurrence at same
- complement event
  A → event, other than A
  A' or $\bar{A}$ or A^c

* Mutually exclusive (or disjoint) events :- if one occur it prevent occurance of the remaining events. it donot occur simultaneously.

- independent events :- occurance 1 event dosent influence the occur of remaining other event

- dependent events :-

— Classical definition / Mathematical approach
  * events are equally likely.
  A → event

P(A) = $\dfrac{\text{Favorable nos of outcomes}}{\text{Total nos of outcomes}}$

$= \dfrac{m}{n}$

* $0 \leq P(A) \leq 1$
  $P(A) = 0$, A is an impossible event.
  $P(A) = 1$, A is a sure event

<u>Axiomatic Approch</u>

1) $P(A) \geq 0$ non negativity
2) $P(S) = 1$ sertainity
3) let $A_1, A_2, A_3 \dots A_n$ are disjoint events

$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$

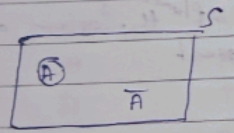
Additivity

<u>rules</u>:

1) complementation

$A \rightarrow$ events

$\bar{A} \rightarrow$ complement



A & $\bar{A}$ are disjoint

$A \cup \bar{A} = S$

$P(A \cup \bar{A}) = P(S)$

i.e $P(A) + P(\bar{A}) = 1$.

or $P(\bar{A}) = 1 - P(A)$

or

$P(A) = 1 - P(\bar{A})$

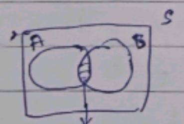
— <u>Note</u>:- Event & its complement are always mutually exclusive

2) Addition rule ($\cup$)

let A & B be two events.

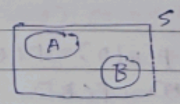
probability of A occurance of <u>atleast</u> (minimum) 1 of 10 is

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$



$A \cap B$

if A & B are disjoint



$A \cap B = \{\phi\}$

$\therefore P(A \cap B) = 0$

$\therefore P(A \cup B) = P(A) + P(B)$

if A, B, C are three events

$P(A \cup B \cup C)$

$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

3)* <u>conditional probability</u> :-

A B — dependence — B | A

or

A B — dependence — A | B

occur A give that occur B

$P(B|A) = \dfrac{P(A \cap B)}{P(A)}$. $P(A) \geq 0$

$P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

$P(A) + P(\bar{A}) = 1$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

or

$P(B|A) = \dfrac{P(A \cap B)}{P(A)}$

* multiplication Theorem.

$A * B \rightarrow$ Two events

$\rightarrow$ simultaneous occurrence

$P(A \cap B) = P(A) \cdot P(B|A)$

or

$= P(B) \ P(A|B)$

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* 5) chatbot

↓

Response

correct 85% — wrong

0.85

3 responses in a row

Resp 1, Resp 2, Resp 3 — correct

respon are independent

A : 1st response is correct

B : 2nd response is correct

C : 3rd response is correct

P [All 3 responses are correct]

$= P(A \cap B \cap C)$

$= P(A) \cdot P(B) \cdot P(C)$

$\therefore$ A, B, C are independent

$= (0.85)(0.85)(0.85)$

$= 0.6141$

6) A robot detect an obstacle 70%.

Robot
↓
detect obstacle
70%

A :- robot detect the obstacle

$\bar{A}$:- robot fails to detect the obstacle

Given $P(A) = 0.70$

$P(\bar{A}) = 1 - P(A)$

$= 1 - 0.70$

$= 0.30$

P (robot fails to detect the obstacle in two consecutive attempts)

= P(1st attempt is failure

&

2nd attempt is failure

= P(1st attempt is failure) * P(2nd attempt is failure)

$= P(\bar{A}) * P(\bar{A})$

$= (0.3) * (0.3)$

$= 0.09$

7) Computer virus

↓

enters the system

email 30% — internet 40%

0.15

A :- virus enter through email

B :- virus enter through internet

$P(A) = 0.30$, $P(B) = 0.40$, ~~P(A∩B)~~

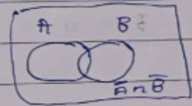
& $P(A \cap B) = 0.15$

P (virus doesn't enter the system)

= P (virus does not enter through email &

virus does not enter through internet)

$= P(\bar{A} \cap \bar{B})$



$\bar{A} \cap \bar{B} = \overline{A \cup B}$

$= 1 - (A \cup B)$

$= P(\overline{A \cup B})$

$= 1 - P(A \cup B)$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= 1 - [0.3 + 0.4 - 0.3 \times 0.4]$

$= 1 - [0.7 - 0.15] = 1 - 0.58 = 0.45$

LP ii) let C_1 correspond to class sail = yes
C_2 correspond to class sail = No

X = (outlook = "rainy", company = "big", sailboat = "small")

we need to maximize $P(X|C_i)\ P(C_i)$ for $i = 1,2$

$P(C_i)$ can be computed on training sample

$n = 10$

yes = 7 $\quad P(sail = yes) = \frac{7}{10} = 0.7$

No = 3 $\quad P(sail = No) = \frac{3}{10} = 0.3$

To compute $P(X|C_i)$ for $i = 1,2$

$P(outlook = \text{"rainy"} \mid sail = yes) = \frac{2}{7}$

$P(outlook = \text{"rainy"} \mid sail = No) = \frac{3}{3} = 1$

$P(company = \text{'big'} \mid sail = yes) = \frac{3}{7}$

$P(company = \text{'big'} \mid sail = No) = \frac{0}{3} = 0$

$P(sailboat = \text{'small'} \mid sail = yes) = \frac{4}{7}$

$P(sailboat = \text{'small'} \mid sail = No) = \frac{1}{3}$

$\therefore P(X \mid sail = yes) = \prod_{k=1}^{n} P(x_k \mid yes)$

$= \left(\frac{2}{7}\right)\left(\frac{3}{7}\right)\left(\frac{4}{7}\right) = \cancel{0.070}\ \frac{24}{343}$

$= 0.06997$

$P(X \mid sail = No) = \prod_{k=1}^{n} P(x_k \mid No)$

$= (1)(0)\left(\frac{1}{3}\right) = 0$

$P(X \mid sail = yes)\ P(sail = yes) = 0.6997 \times 0.7$

$= 0.0489$

$P(X \mid sail = No) \cdot P(sail = No) = 0 \times 0.3 = 0$